



Due data: 01/03/2026.

1. [2] Caesar Cipher (Short Text)

Ciphertext: dvvkzcefssprkkve

Tasks:

1. Perform brute-force decryption by trying all possible shifts (0–25).
2. Identify the meaningful plaintext.
3. Verify your result with frequency analysis (most frequent ciphertext letter vs. expected English distribution).

2. [3] Caesar Cipher (Longer Text)

Ciphertext: ymnxhtzwxjfnrxytuwtanijdtzbnymijyfnqjipstbqiljtknrtwyfsyyjhmstqtlnjxfsifuuqnh
fyntsymfyfwjzjinsymjnsyjsyzytymjgwtfsfyzwjtkymnxknjqiyjmjhtzwxjhtajwxtsqdx
jqjhyjiytunhxxthzxxnslknwxytsxtrjfiashjiytunhxnnsyjsyzyjhmstqtlnjxjlbwjqjxx
qfsxrtgnqjnsyjsyjrzyqnfhxfysymjsfxjqjhyntstkhzwwjsyfsisjcyjsjwfyntsfuuqnhfyn
tsxfsixjwanhxxjluunuyaatanudtznqqjfwsmbymjnsyjsybtwpxfsimtbxjwanhxxfsifuuqn
hfyntsfwjuwtanijiytzxjwxtkymjnsyjsyymnxpstbqiljbnqqmqjqudtz

Tasks:

1. Count letter frequencies in the ciphertext.
2. Hypothesize the most likely key by aligning the most common ciphertext letter with 'E'.
3. Decrypt the ciphertext with this key.
4. Show frequency charts (ciphertext vs English).
5. Submit the recovered plaintext.

3. [5] Monoalphabetic Substitution Cipher

Ciphertext: ziolegxkltdqodlzfzkgfzgrxetngxzgzihtkfoehstlqfrzteifoxtlgytlxkofuegdhxxztklqfregd
hxztkftzvgkalvoziygexlzfztkftzltexkzozniztegxkltoitytezoctsnlhsozofzgzvghqkzlyo
klzofzkgfzgrxetfztkftzltexkzozniztegxkltoitytezoctsnlhsozofzgzvghqkzlyo
lvgkaturltklqrouozqsloufzqzxtlqfzltexkzozniztegxkltoitytezoctsnlhsozofzgzvghqkzlyo
gegslqsugkozidqfztkftzltexkzozniztegxkltoitytezoctsnlhsozofzgzvghqkzlyo
lhgezlguegdhxxztklqfztkftzltexkzozniztegxkltoitytezoctsnlhsozofzgzvghqkzlyo
afgvstrutvossitshngxofztkftzltexkzozniztegxkltoitytezoctsnlhsozofzgzvghqkzlyo
qlvtssqlwxosofztkftzltexkzozniztegxkltoitytezoctsnlhsozofzgzvghqkzlyo

Tasks:

1. Compute single-letter frequencies.
2. Compare with English letter frequencies (E, T, A, O most common).
3. Analyze common digrams/trigrams ('TH', 'HE', 'ING').
4. Iteratively replace likely mappings and refine until meaningful plaintext is obtained.
5. Document all steps and reasoning.



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BS – CS/SE/AI

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CY2101 Information Security

Total Marks: 10

Assignment 1

Deliverables

- An Assignment report including:
- Frequency analysis graphs/tables
- Clearly mention Step-by-step decryption attempts.
- Final plaintexts for all parts.